

AI Engineer

1. What is an AI Engineer?

An AI Engineer is a software engineer who builds products on top of AI models, rather than building the models themselves.

The work is integration, not research: choosing a model, giving it the right context, handling what it returns, and making the result reliable enough to ship.

The models are bought, not made. An AI Engineer calls a model the way an earlier generation called a database, and the skill is in everything around the call.

2. Roles and responsibilities

Responsibility	What it involves
Choosing a model	Matching capability, latency and cost to the task
Designing prompts	Getting reliable output from an unreliable component
Supplying context	Retrieving the right data and fitting it in the window
Handling output	Validating, parsing, and deciding what to do when it is wrong
Evaluating	Measuring quality, because tests alone cannot
Controlling cost	Tokens are billed, so design decisions are spending decisions
Guarding	Preventing prompt injection, leaks and unsafe output

Evaluation is the part that is genuinely new. Ordinary code is right or wrong; a model's answer is better or worse, so the job includes deciding what "good enough" means and measuring it.

3. Impact on product development

Before	With a model in the product
Features are specified exactly	Behavior is guided, not specified
Output is deterministic	The same input can give different output
Testing proves correctness	Evaluation estimates quality
Cost is servers	Cost is per request, and grows with use
A bug is a defect	A wrong answer may be within normal behavior

Products can now do things nobody could specify. Summarizing, drafting, classifying open-ended text and answering in natural language were not features that could be written down as rules, and that is what changed.

The price is that certainty is gone. A feature built on a model needs a plan for being wrong, and that plan is part of the design rather than an afterthought.

4. AI Engineer against ML Engineer

	AI Engineer	ML Engineer
Builds	Products that use models	The models, and the systems that train them
Starts from	A model that already exists	Data
Main skills	Software engineering, prompting, retrieval	Statistics, modeling, training pipelines
Typical output	A feature in an application	A trained model and a serving endpoint
Cares about	Latency, cost, correctness of the answer	Accuracy, drift, training cost

The dividing line is whether model weights change. An ML Engineer produces and improves them. An AI Engineer takes them as given and builds around them.

5. AI Engineer against Data Scientist

	AI Engineer	Data Scientist
Goal	Ship a working feature	Answer a question about the data
Output	Production code	An analysis, a model, a recommendation
Audience	Users	Decision makers
Ends when	It runs reliably for everyone	The question is answered

A Data Scientist explains; an AI Engineer ships. The two overlap in tooling and almost nowhere in what counts as finished.

6. How does AI fit into modern software development?

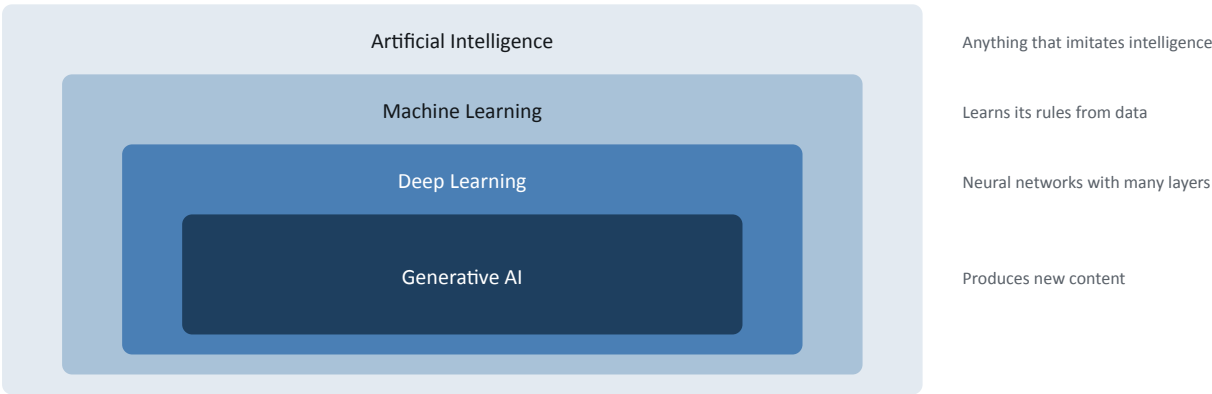
As one more remote dependency, with unusual properties.

Like any dependency	Unlike any dependency
Called over the network	Answers differently to the same input
Can be slow, or unavailable	Can be confidently wrong
Costs money per call	Cost varies with the length of the input
Needs timeouts and retries	Needs evaluation, not only tests

It sits behind the same patterns everything else does. Timeouts, retries, circuit breakers, caching and observability all apply, and the architecture around a model looks much like the architecture around a slow third-party API.

7. AI, Machine Learning, Deep Learning and Generative AI

These four are nested, not alternatives.



Each one is a way of doing the one outside it. Generative AI is deep learning, which is machine learning, which is artificial intelligence.

Term	Means
Artificial Intelligence	Any system that performs tasks needing intelligence
Machine Learning	AI where the rules are learned from data, not written
Deep Learning	Machine learning using neural networks of many layers
Generative AI	Deep learning that produces new content rather than a label

8. What is Machine Learning?

Machine Learning is the practice of teaching a program the rules from examples, instead of writing the rules.

Traditional rules + data → answers

Machine Learning data + answers → rules

That reversal is the whole idea. Nobody can write the rules for recognizing a cat in a photograph; a model can learn them from enough labeled photographs.

9. The types of Machine Learning

Type	Learns from	Example
Supervised	Labeled examples	Spam or not spam, from marked emails
Unsupervised	Unlabeled data	Grouping customers by behavior
Reinforcement	Rewards for actions taken	A model learning to play a game

Supervised learning is most of what is in production, because most business problems come with historical answers attached.

A fourth is worth knowing because it built the current models: **self-supervised** learning, where the labels come from the data itself. Hiding the next word in a sentence and asking the model to predict it needs no human labeling at all, which is why it scaled.

10. What is Deep Learning?

Deep Learning is machine learning using neural networks with many layers.

"Deep" refers to the number of layers, and the depth is what lets each layer build on the one before it: edges, then shapes, then objects.

It replaced feature engineering. Earlier machine learning needed a person to decide which properties of the data mattered. A deep network works that out itself, given enough data and computation.

11. What is a neural network?

A neural network is a set of connected units arranged in layers, where each connection has a weight that is adjusted during training.

input → layer → layer → layer → output

Each unit takes numbers in, multiplies them by its weights, adds them up, applies a simple function, and passes the result on.

Training means adjusting the weights. The network makes a prediction, the error is measured, and every weight is nudged in the direction that would have reduced it. Repeated over enough examples, that is learning.

A trained model is just its weights. Downloading a model downloads numbers, and running it is arithmetic.

The pieces together

Term	In one line
AI Engineer	Builds products on models somebody else trained
Against an ML Engineer	One ships features on a model, the other produces the model
Against a Data Scientist	One builds software, the other answers questions with data
Artificial Intelligence	Any system doing work that needs intelligence
Machine Learning	Rules learned from data instead of written down
Supervised, unsupervised, reinforcement	Labeled examples, structure alone, reward
Deep Learning	Machine learning with many-layered neural networks
Generative AI	The layer of deep learning that produces content
Neural network	Weights adjusted until the output matches the examples

For example, a team adding a "summarize this thread" button. No one on it trains anything. The work is choosing a model, deciding what text to send, writing and versioning the instruction, setting a limit on the answer length, deciding what to show when the model is unavailable or wrong, and building the set of threads that says whether a change to any of it made things better. That list is the job, and none of it is machine learning in the older sense.

The role exists because a capability that used to need a research team now needs an integration. What separates the people doing it well is not knowing how the model was trained. It is knowing that the model is fixed, that everything around it is ours, and that the parts around it are where products are won or lost.